

SONiX 32-Bit Cortex-M0 Micro-Controller

OPA Example Code

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AMENDENT HISTORY

Version	Date	Description
1.0	2024/11/29	1. First version.

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1 OVERVIEW

This document provides our customers the C program examples of OPA.

2 FEATURE

- 3 * OPA.
- 1X~32X PGA.
- Use ARM-compiler V6.12

3 HW

- The Starter kit board of vendor desired MCU.

4 EXAMPLE CODE

1. Set OPA_Enable function in main.c, the related codes are provided in main function.

```

52 int main(void)
53 {
54     //User can configure System Clock with Configuration Wizard in system_SN32F400.c
55     SystemInit();
56     SystemCoreClockUpdate();           //Must call for SN32F400, Please do NOT remove!!!
57
58     //Note: User can refer to ClockSwitch sample code to switch various HCLK if needed.
59
60     PFPA_Init();                       //User shall set PFPA if used, do NOT remove!!!
61
62     //1. User SHALL define PKG on demand.
63     //2. User SHALL set the status of the GPIO which are NOT pin-out to input pull-up.
64     NotPinOut_GPIO_init();
65
66     //-----
67     //User Code starts HERE!!!
68     OPA_Enable();

```

2. If we want to enable OPA clock, call __OPA_ENABLE_PCLK in OPA_Enable function.

```

38 /******
39 * Function      : OPA_Enable
40 * Description   : Enable PCLK for OPA
41 * Input        : None
42 * Output       : None
43 * Return       : None
44 * Note        : None
45 *****/
46 void OPA_Enable(void)
47 {
48     __OPA_ENABLE_PCLK;
49 }

```

3. In OPA_PGA_Test function, and select OPAx and PGA_GAIN value.

```
37 //User define
38 #define PGA_GAIN      0x1F

70 //OPA PGA mode example
71 OPA_PGA_Test(OPA0_ENABLE,PGA_GAIN);
72
73 while (1){}
```

4. Enter OPA_PGA_Test function, and setting OPA_PGACTRL / PGACTRL2 / CTRL registers.

```
64 /*
65 * Function      : OPA_PGA_Test
66 * Description   : OPA_PGA_Test
67 * Input        : number (OPA0~2)
68 *               : gain (setting value + 1), setting value = 0x0~0x1F(1X~32X)
69 * Output       : None
70 * Return       : None
71 * Note        : None
72 */
73 void OPA_PGA_Test(uint8_t number, uint8_t gain)
74 {
75     SN_OPA->PGACTRL_b.VHALF_EN = 1; //0: Disable, 1: Enable
76
77     if(number == OPA0_ENABLE)
78     {
79         SN_OPA->PGACTRL |= (mskOP0PS_DIS | mskOP0NS_EN | mskOP0PRS_10K | mskOP0NRS_10K);
80         SN_OPA->PGACTRL_b.OP0GS = gain;
81         SN_OPA->CTRL |= (mskOP0EN_EN | mskOP0_SEL_OP00);
82     }
83     if(number == OPA1_ENABLE)
84     {
85         SN_OPA->PGACTRL |= (mskOP1PS_DIS | mskOP1NS_EN | mskOP1PRS_10K | mskOP1NRS_10K);
86         SN_OPA->PGACTRL_b.OP1GS = gain;
87         SN_OPA->CTRL |= (mskOP1EN_EN | mskOP1_SEL_OP10);
88     }
89     if(number == OPA2_ENABLE)
90     {
91         SN_OPA->PGACTRL2 |= (mskOP2PS_DIS | mskOP2NS_EN | mskOP2PRS_10K | mskOP2NRS_10K);
92         SN_OPA->PGACTRL2_b.OP2GS = gain;
93         SN_OPA->CTRL |= (mskOP2EN_EN | mskOP2_SEL_OP20);
94     }
95 }
```

5 RESULT

The OPA PGA mode is noninverting amplifier, and V+ is input voltage, V- is tie GND. In this case, input voltage is 0.141V, PGA gain value is 0x1F (gain=32x), because OPA is noninverting amplifier, so real Av gain is 33x. Final theoretical Vo is $\sim 4.653V$.

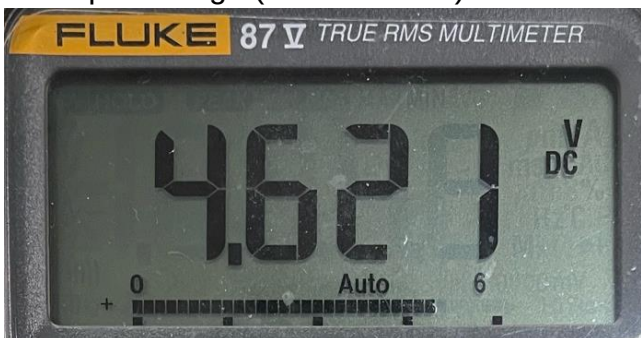
- 5V MCU VDD (OPA power)



- 0.141V Input voltage (OPA V+)



- Output voltage (Real OPA Vo)



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